

## CYBER SECURITY COURSE

# Network Service Enumeration

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## Professional Case Report 11

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| <b>Report date</b>     | 24. septembar 2026                |
| <b>Case ID</b>         | CSC-2026-11                       |
| <b>Discipline</b>      | Network assessment                |
| <b>Overall risk</b>    | High                              |
| <b>Evidence status</b> | Laboratory methodology documented |

### Purpose

Produce a reliable service inventory and understand how scan type affects visibility and network evidence.

### Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

## Executive Summary

A controlled network assessment compared Netcat, Nmap and NSE techniques for host discovery, TCP and UDP scanning, service identification and focused vulnerability checks. The assessment method separates discovery from validation. That approach reduces false assumptions and produces findings that can be tied to specific services and controls.

## Case Profile

|             |                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scenario    | A controlled network assessment compared Netcat, Nmap and NSE techniques for host discovery, TCP and UDP scanning, service identification and focused vulnerability checks. |
| Environment | Kali Linux scanning isolated Windows and Linux training targets.                                                                                                            |
| Tools       | Netcat, Nmap, Nmap Scripting Engine, iptables counters, tcpdump                                                                                                             |

## Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Performed narrow TCP and UDP Netcat scans.
2. Compared Nmap default, SYN, connect and UDP scan behavior.
3. Measured traffic through iptables counters.
4. Used NSE for SMB discovery, DNS checks and vulnerability-oriented validation.

## Technical Findings

| ID  | Finding                                 | Risk   | Evidence and analysis                                                                            |
|-----|-----------------------------------------|--------|--------------------------------------------------------------------------------------------------|
| F01 | Scan methods produce different evidence | Medium | SYN, connect and UDP scans differ in privilege requirements, accuracy and log visibility.        |
| F02 | Service enumeration is required         | High   | An open port becomes actionable only after protocol, version and configuration are identified.   |
| F03 | NSE supports focused validation         | High   | Targeted scripts can confirm NFS, SMB and known-vulnerability conditions without a full scanner. |

## Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
nmap -sS <target_ip>
nmap -sT <target_ip>
sudo nmap -sU <target_ip>
nmap <target_ip> --script=smb-os-discovery
```

```
nmap --script vuln <target_ip>
```

## Risk Assessment

| Risk             | Level  | Operational impact                                        |
|------------------|--------|-----------------------------------------------------------|
| Exposed services | High   | Unnecessary listeners increase attack surface             |
| UDP blind spots  | Medium | Slow or incomplete UDP review can miss important services |

## Recommendations

1. Maintain an approved service inventory.
2. Close unnecessary TCP and UDP ports.
3. Segment management services from user networks.
4. Use focused NSE validation after broad discovery.

## Knowledge Demonstrated

- TCP and UDP scanning
- Service fingerprinting
- Packet-level reasoning
- NSE scripting

## Conclusion

The assessment method separates discovery from validation. That approach reduces false assumptions and produces findings that can be tied to specific services and controls.

## Hands On Lab Practical 11

Completed as part of the authorized cyber security course practical work.